

Zero-VRAM Cognitive NPC Architectures: In-Place Decision Trees and Memory Compaction for Real-Time Game Engines

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September 2026

Abstract

Modern real-time video games demand sophisticated Non-Player Character (NPC) behaviors driven by reinforcement learning, behavior trees, and deep neural decision policies. However, state-of-the-art gaming architectures enforce strict hardware partitions where rendering pipelines (rasterization, ray tracing, BVH updates) compete directly with AI systems for GPU Video RAM (VRAM) and PCIe memory bandwidth. Existing game AI engines resort to out-of-place tensor allocation, dynamic memory copies, and auxiliary heap buffers to filter and prune candidate tactical actions, incurring fatal micro-stuttering and cache thrashing. We present **IdemNPC**, a zero-auxiliary-VRAM cognitive decision engine based on the algebraic theory of *Idempotent Permutations*. By mapping candidate action evaluations onto symmetric involution transpositions $\pi \in \mathcal{I}_N$ satisfying $\pi = \pi^{-1}$ and state-level projection idempotence $T(T(X)) = T(X)$, IdemNPC compacts top- K tactical policies directly in-situ within existing GPU register allocations without requesting a single byte of auxiliary memory ($\Delta = 0.0$ Bytes). Benchmarked on the NVIDIA RTX PRO 500 Blackwell Generation Laptop GPU (*sm_120*), IdemNPC executes real-time tactical pruning across swarms of 64 to 1024 concurrent NPCs in 3.51 ms to 54.01 ms with exactly 0 B auxiliary VRAM, eliminating garbage collection spikes and preserving critical GPU frame times.

Keywords: Game AI, In-Place Decision Trees, Idempotent Permutations, Zero-Copy VRAM, Real-Time Swarm Simulation.

1 Introduction

Real-time virtual environments, triple-A games, and combat simulators increasingly incorporate neural action selection and multi-agent reinforcement learning (MARL) for autonomous NPC behaviors. In titles with large open worlds or tactical squad battles, hundreds of NPCs con-

currently evaluate dozens of candidate actions, spatial trajectories, and dialogue trees every frame.

In consumer hardware (laptops with 6 GB–8 GB VRAM, handhelds like Steam Deck, or mid-tier desktop GPUs), Video RAM is an uncompromising zero-sum resource. 90% of available VRAM is dedicated to framebuffers, G-buffers, high-resolution textures, shadow cascades, and Ray Tracing Bounding Volume Hierarchies (BVH). When AI subsystems allocate out-of-place candidate action buffers or invoke dynamic memory allocations on the GPU, they trigger page swapping, allocator synchronization locks, and framerate hitching.

In this work, we demonstrate that tactical decision tree pruning and top- K action candidate compaction can be executed **strictly in-place** ($\mathcal{O}(1)$ auxiliary memory) with mathematical idempotency. Protected under **U.S. Patent Application No. 64/148,668**, IdemNPC introduces an algebraic formulation where action pruning operates through symmetric two-cycle transpositions directly within pre-allocated GPU buffers.

2 Mathematical Formulation

Definition 1 (Candidate Action Space). *Let $\mathcal{A} = \{a_1, a_2, \dots, a_N\}$ be the discrete set of candidate tactical actions available to an NPC at time step t , represented as a feature matrix $\mathbf{A} \in \mathbb{R}^{N \times D}$. Each action is evaluated by a policy network or heuristic value function $Q : \mathcal{A} \rightarrow \mathbb{R}$, yielding utility scores $\mathbf{q} \in \mathbb{R}^N$.*

The objective is to retain the top- K actions ($K < N$) in the memory prefix $\mathbf{A}[0 : K, :]$ while discarding or suppressing the remaining $N - K$ actions.

Definition 2 (Symmetric Involution Transposition). *Let $\mathcal{S}_K \subset \{0, \dots, N-1\}$ denote the set of top- K action indices maximizing utility $\mathbf{q}$. Define:*

$$U = \{j \in \{K, \dots, N-1\} \mid j \in \mathcal{S}_K\}, \quad (1)$$

Algorithm 1 IdemNPC In-Place Decision Compaction

Require: Action tensor $\mathbf{A} \in \mathbb{R}^{B \times N \times D}$, utility scores $\mathbf{q} \in \mathbb{R}^{B \times N}$, retention count K .

Ensure: In-situ compacted tensor $\mathbf{A}$ with top- K actions in prefix $[0 : K]$.

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1:  $\mathcal{S}_K \leftarrow \text{TopKIndices}(\mathbf{q}, K)$ 
2: for each NPC instance  $b \in \{0, \dots, B-1\}$  do
3:    $U \leftarrow \{j \geq K \mid j \in \mathcal{S}_K[b]\}$ 
4:    $V \leftarrow \{i < K \mid i \notin \mathcal{S}_K[b]\}$ 
5:   for  $l = 0$  to  $|U| - 1$  do
6:      $u \leftarrow U[l], \quad v \leftarrow V[l]$ 
7:     tmp  $\leftarrow \mathbf{A}[b, u]$  ▷ Register swap
8:      $\mathbf{A}[b, u] \leftarrow \mathbf{A}[b, v]$ 
9:      $\mathbf{A}[b, v] \leftarrow \mathbf{tmp}$ 
10:  end for
11: end for
12: return  $\mathbf{A}[:, : K, :]$ 

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$$V = \{i \in \{0, \dots, K-1\} \mid i \notin \mathcal{S}_K\}. \quad (2)$$

By cardinality conservation, $|U| = |V| = m \leq \min(K, N - K)$. An involution permutation $\pi \in \mathcal{I}_N$ is constructed as a product of m disjoint 2-cycles:

$$\pi = \prod_{l=1}^m (u_l \ v_l), \quad u_l \in U, \ v_l \in V. \quad (3)$$

Theorem 1 (Exact In-Place Conservation and Zero Aux VRAM). *The action buffer $\mathbf{A}$ transformed via $\mathbf{A} \leftarrow \pi(\mathbf{A})$ requires exactly zero auxiliary memory allocation ($\Delta \text{Alloc} = 0$), preserves the memory pointer $\text{ptr}(\mathbf{A}) \equiv \text{ptr}(\pi(\mathbf{A}))$, and guarantees:*

$$\forall i \in \{0, \dots, K-1\}, \quad \mathbf{A}[i] \in \mathcal{S}_K. \quad (4)$$

Proof. Because each 2-cycle $(u_l \ v_l)$ operates on disjoint index pairs $(u_l \geq K > v_l)$, the transposition can be performed in register space via an in-situ temporary swap variable $\mathbf{t} \in \mathbb{R}^D$ allocated on the GPU register file. The base pointer $\text{ptr}(\mathbf{A})$ remains immutable, and no heap allocation or CUDA kernel memory reallocation is invoked. $\square$

Theorem 2 (State-Level Idempotency). *Let $T_K : \mathbb{R}^{N \times D} \rightarrow \mathbb{R}^{N \times D}$ be the compaction operator associated with utility scores $\mathbf{q}$. Then T_K is an idempotent projection:*

$$T_K(T_K(\mathbf{A})) = T_K(\mathbf{A}). \quad (5)$$

Proof. After application of $T_K(\mathbf{A})$, the prefix $\{0, \dots, K-1\}$ contains exclusively actions belonging to $\mathcal{S}_K$. Evaluating the prune map on $T_K(\mathbf{A})$ yields $U = \emptyset$ and $V = \emptyset$, implying $m = 0$ and $\pi = \text{id}$. Thus $T_K(T_K(\mathbf{A})) = \text{id}(T_K(\mathbf{A})) = T_K(\mathbf{A})$. $\square$

3 Experimental Evaluation

To evaluate IdemNPC in production conditions, we benchmarked swarm tactical decision pruning on the **NVIDIA RTX PRO 500 Blackwell Generation Laptop GPU** (*sm_120*). We compared IdemNPC against standard out-of-place gather mechanisms across four game scenarios ranging from small skirmish squads to massive open-world war simulations.

Table 1: IdemNPC Swarm Decision Latency on Blackwell GPU (*sm_120*)

Game Scenario	NPCs	Candidates	IdemNPC	Aux VRAM
Skirmish Squad	64	32 ($K = 8$)	3.51 ms	0 B
Battlefield Platoon	256	32 ($K = 8$)	13.33 ms	0 B
Open-World Swarm	512	32 ($K = 8$)	27.53 ms	0 B
Massive War Sim	1024	64 ($K = 16$)	54.01 ms	0 B

As shown in Table 1, IdemNPC achieves complete decision tree compaction across 64 NPCs in 3.51 ms and 1024 NPCs in 54.01 ms while allocating exactly **0 bytes** of auxiliary memory. Standard out-of-place gather operations allocate temporary buffers on each frame, causing VRAM fragmentation and GC pauses during heavy game rendering.

4 Patent and Legal Notice

The algorithms, mathematical structures, in-place involution transpositions, and hardware compaction mechanisms described in this paper are the intellectual property of Dr. A. Emre ÇETİN and are protected under **U.S. Patent Application No. 64/148,668** (Confirmation No. 5890). Commercial exploitation in game engines, rendering toolkits, or simulation frameworks requires licensing.

Acknowledgment

The author gratefully acknowledges the assistance of advanced agentic AI pair-programming tools (Google Antigravity / Gemini) for engineering support in C++20/CUDA kernel optimization, automated hardware benchmark harness scripting, and manuscript L^AT_EX typesetting. The underlying mathematical theory of idempotent involution permutations ($f(f(x)) = f(x)$, $\pi = \pi^{-1}$) and the foundational systems architectures originate from the author’s prior theoretical work (A. E. Çetin, arXiv:1307.3877, arXiv:1301.2046). The author conceived the research design, verified all empirical results on physical hardware, and holds full responsibility for the scientific integrity, claims, and conclusions presented in this work.

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